Tutorial: World Bank Open Data in LibreOffice Calc

A hands-on walkthrough of every function and feature in the World Bank Open Data Calc add-in. No API key, no signup — just install and start typing formulas. Screenshots aren't included here, but every formula below is copy-pasteable into a real sheet; the sample outputs were captured from a live run against the real API on 2026-07-16; the World Bank periodically revises GDP and similar figures, so your numbers may differ slightly, and "most recent year" will keep advancing as new years are published.

What you'll build

By the end of this tutorial you'll have a working sheet that:

- looks up a single indicator value for a specific year,
- pulls a full time series and charts it,
- compares two countries side by side,
- labels indicator codes automatically,
- and gracefully survives typos and network hiccups.

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0. Install it

If you haven't already:

```
"$LO_HOME/program/unopkg" add --force build/WorldBank.oxt
```

(No build tools? Download the prebuilt `WorldBank.oxt` from the releases page instead — see the README.) Restart LibreOffice, then open a new Calc document.

Confirm it's registered before going further — type this in any cell:

```
=WBMETA("NY.GDP.MKTP.CD")
```

If you see `#NAME?`, the extension didn't register; check `unopkg list` and see INSTALL.md. Anything else (including `#FETCHING`) means you're good to continue.

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1. Your first lookup: `WBVALUE`

Every function needs two things at minimum: a **country code** and an **indicator code**. Countries are ISO2 (`"US"`), ISO3 (`"USA"`), or a World Bank aggregate (`"WLD"` for World). Indicators are World Bank's dotted codes — `NY.GDP.MKTP.CD` is GDP in current US dollars. (More codes in the README's table, or look one up live with `WBMETA` — see Part 5.)

In cell `A1`, type:

```
=WBVALUE("US"; "NY.GDP.MKTP.CD"; 2020)
```

Note the **semicolons** — that's Calc's argument separator, not commas.

The first time you enter this formula, don't be surprised if you see:

```
#FETCHING
```

That's not an error — it means the add-in just kicked off a background request and handed the cell a placeholder immediately, rather than freezing your spreadsheet while it waits on the network. Press **Ctrl+Shift+F9** (recalculate hard) a second or two later, and it resolves:

```
21375281000000
```

That's the United States' 2020 GDP in current dollars.

Omit the year for "most recent"

Drop the third argument entirely and you get whatever year the World Bank has most recently published:

```
=WBVALUE("US"; "NY.GDP.MKTP.CD")
```

→ resolves near-instantly (see "One fetch, three functions" below) to the latest available figure — e.g. `30769700000000` for 2025, as of this writing.

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2. Understanding the three sentinels

Besides real numbers, any `WB*` function can show one of three short-lived placeholder strings instead of throwing an error or freezing the UI:

You see	What it means	What to do
`#FETCHING`	First request for this data; a background fetch just started.	Wait a second or two, recalculate (F9 or Ctrl+Shift+F9).
`#NOT_FOUND`	The request reached the API, but nothing matched — e.g. every value in range is null.	Try a different year/range, or double-check the country/indicator pairing.
`#ERR`	The fetch failed persistently — usually an invalid indicator or country code.	Call `WBLASTERROR()` (Part 6) for the detail message.

Try triggering `#ERR` on purpose — type a made-up indicator code:

```
=WBVALUE("US"; "BOGUS.INDICATOR.XYZ")
```

You'll get `#ERR`. That's the add-in correctly detecting that the World Bank API rejected the code, rather than silently returning a wrong number. Keep this cell around — you'll use it in Part 6.

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3. A time series: `WBSERIES`

`WBSERIES` returns a whole table, so it needs an **array formula**: select a *range* of cells (not just one), type the formula, then press **Ctrl+Shift+Enter** instead of plain Enter (LibreOffice has no automatic spill, so this step is required every time). If you forget, you'll only see the top-left value.

Select `A3:B12` (10 rows, 2 columns), type:

```
=WBSERIES("US"; "NY.GDP.MKTP.CD"; 2015; 2020)
```

and confirm with **Ctrl+Shift+Enter**. You'll get 6 rows (2015 through 2020 inclusive), sorted ascending by year:

year	value
2015	18295019000000
2016	18804913000000
2017	19612102000000
2018	20656516000000
2019	21539982000000
2020	21375281000000

Notice row 2020's value matches the `WBVALUE` lookup from Part 1 — same underlying data, two different views.

The full history, for free

Omit both year bounds and you get *every* year the World Bank has for that country/indicator — currently back to 1960 for GDP:

```
=WBSERIES("US"; "NY.GDP.MKTP.CD")
```

Select a generous range (at least ~70 rows for a long-running annual indicator) before entering it. This is a good formula to pair with **Insert** → **Chart** for an instant line chart of six decades of GDP.

Null years show up blank, never as zero

If a country/indicator combination has gaps (a young country reporting a long-running indicator, for instance), those years appear in the output with an **empty** value cell — never `0`. A `0` in a WBSERIES row always means the World Bank actually reported zero, not "no data."

Multiple countries at once

Join country codes with a semicolon *inside the string* (this is separate from Calc's own argument-separating semicolons):

```
=WBSERIES("US;GB"; "NY.GDP.MKTP.CD"; 2022; 2023)
```

Select a **3-column** range this time (`A;B;C`) — WBSERIES automatically adds a leading `country` column whenever you name more than one country:

country	year	value
GB	2022	3181244350465.41
US	2022	26054614000000
GB	2023	3420796653789.08
US	2023	*(next row)*

This is the fastest way to build a side-by-side country comparison without four separate formulas.

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4. Dashboard tiles: `WBLATEST`

`WBSERIES` and repeated `WBVALUE` calls are great for tables, but a dashboard often wants just "the current number and what year it's from" as a compact tile. That's `WBLATEST` — also an array formula, but always exactly one row, two columns:

Select `A20:B20`, type:

```
=WBLATEST("US"; "NY.GDP.MKTP.CD")
```

confirm with Ctrl+Shift+Enter, and you get:

value	year
30769700000000	2025

One fetch, three functions

Here's a detail worth knowing: `WBVALUE`, `WBSERIES`, and `WBLATEST` for the **same country + indicator** all share one cached fetch. The World Bank API hands back a country's entire history in a single request, so the add-in pulls it once and answers "value in year Y," "most recent value," and "the full series" all from that same cached data. Try it: after any of the formulas above have resolved for `("US", "NY.GDP.MKTP.CD")`, enter another one for the same pair — it resolves instantly, no `#FETCHING` at all, because there's nothing left to fetch.

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5. Labeling: `WBMETA`

Indicator codes like `NY.GDP.MKTP.CD` aren't self-explanatory in a shared sheet. `WBMETA` turns a code into its human-readable name:

```
=WBMETA("NY.GDP.MKTP.CD")
```

→ `GDP (current US$)`

This shines when you're building a small reference table. Put codes down a column and labels next to them:

A	B
`NY.GDP.MKTP.CD`	`=WBMETA(A1)` → GDP (current US$)
`SP.POP.TOTL`	`=WBMETA(A2)` → Population, total
`FP.CPI.TOTL.ZG`	`=WBMETA(A3)` → Inflation, consumer prices (annual %)

Then reference column A (the code) in your `WBVALUE` / `WBSERIES` formulas and column B (the label) as your chart/table headers — change the code in one place, and both the data and its label update together.

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6. Diagnostics: `WBLASTERROR` and `WBCACHECLEAR`

Remember the `#ERR` cell from Part 2? Now find out *why* it failed:

```
=WBLASTERROR()
```

→ `World Bank API error: The provided parameter value is not valid`

That's the World Bank API's own rejection message for the bogus indicator code, surfaced verbatim rather than a generic "something went wrong." `WBLASTERROR()` always reflects the *most recent* failure across every `WB*` function in the session — handy as a single diagnostic cell near the top of a sheet full of formulas.

If you want to force a completely fresh pull of everything (e.g. you suspect World Bank has revised a figure since your session started, and don't want to wait out the 24-hour cache TTL):

```
=WBCACHECLEAR()
```

→ returns the number of cache entries cleared. One quirk worth knowing: LibreOffice's formula engine doesn't guarantee a cell's function runs *exactly* once per recalculation, so this count is best-effort — treat it as "yes, something got cleared," not an exact audit number. Every formula you recalculate afterward will show `#FETCHING` again as it refetches.

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7. Put it together: a two-country GDP comparison

As a capstone, build a small comparison sheet from scratch:

1. **A1**: `Country` **B1**: `GDP label` **C1**: `2023 GDP` **D1**: `Latest GDP` **E1**: `Latest year`
2. **A2**: `US` **A3**: `GB` **A4**: `DE`
3. **B2**: `=WBMETA("NY.GDP.MKTP.CD")` (only need this once — copy down if you like, it's the same for every row)
4. **C2**: `=WBVALUE(A2; "NY.GDP.MKTP.CD"; 2023)` — fill down to C3, C4
5. Select **D2:E2**, enter `=WBLATEST(A2; "NY.GDP.MKTP.CD")` as an array formula (Ctrl+Shift+Enter), then repeat for rows 3 and 4

Recalculate (Ctrl+Shift+F9) once or twice to clear any `#FETCHING` placeholders, and you have a live, three-country GDP comparison built from five formulas — each country's three columns (2023 value, latest value, latest year) sharing one cached fetch per country, so recalculating this sheet repeatedly costs nothing extra.

Quick reference

Function	Entry mode	Purpose
`WBVALUE(country; indicator; [year])`	normal cell	one number
`WBSERIES(country; indicator; [start]; [end])`	**array formula**	a table of years
`WBLATEST(country; indicator)`	**array formula**	value + year, one row
`WBMETA(indicator)`	normal cell	indicator's name
`WBLASTERROR()`	normal cell	last failure's detail
`WBCACHECLEAR()`	normal cell	force a fresh refetch

Array formula reminder: select the *whole* output range first, type the formula, then **Ctrl+Shift+Enter** (not plain Enter).

Sentinel reminder: `#FETCHING` → wait and recalculate. `#NOT_FOUND` → no data for that combination. `#ERR` → check `WBLASTERROR()`.

Where to go next

- Full function signatures and argument tables: README.md
- Build-from-source and platform-specific install steps: INSTALL.md
- Browse thousands more indicator codes: https://data.worldbank.org/indicator
- A pre-built example workbook with every function already resolved: `test/worldbank_demo.ods`